Fundamental Issues of Statistics

Introduction to Statistics

- What do you think of when you see/hear the word **"Statistics"**? The majority of people immediately think of numerical facts, data, graphs and tables. But not only do statisticians collect, classify and tabulate data, they also analyze data in order to make generalizations and decisions.

Why study statistics?

- Everyone comes in contact with statistics in everyday life.
- People should understand reports in newspapers, magazine and journals.
- People should be able to question the statistics they read, and not blindly accept this as proven fact.

- Many areas of study use statistics, such as; psychology, sociology, business, biology, government, engineering, science education and even areas such as history, language and the arts.

- **Statistics** is the science of collecting, organizing, summarizing, and analyzing data to draw conclusions or answer questions. It also provides a measure of confidence in any conclusions.

- **Statistics** is the discipline that concerns the collection, organization, analysis, interpretation, and presentation of data.

Two types of statistics:

- **Descriptive statistics**: the use of numbers to summarize information which is known about some population. [collecting, organizing and summarizing the data]

  Descriptive Statistics is the branch of statistics that focuses on collecting, summarizing, andpresenting a set of data.

- **Inferential statistics**: the use of numbers related to a random sample from a population to give numerical information about the population itself. [analyzing the data to draw conclusions or answer questions about the population]

  Inferential Statistics is the branch of statistics that analyzes sample data to draw conclusionsabout a population.

- **Probability** is very important in inferential statistics; it's related to the risk of making an error.

- **Variables** are characteristics of the individuals or things being studied. Variable is a characteristic of an individual that will be analyzed using statistics.

Fundamental Issues of Statistics

Two types of variables:

- Qualitative or categorical variable – classification based on some attribute or characteristic of the individual (non-numerical).

  Categorical (qualitative) variables have values that can only be placed into categories, such as "yes" and "no"; major; architectural style; etc. Example: hair color, eye color, gender, race, ethnicity

- Quantitative or numerical variable – provides numerical measures of individuals.

  Numerical (quantitative) variables have values that represent quantities.

Two types of quantitative variables:

- ✓ **Discrete (something that can be counted)** – has either a finite number of possible values or a countable number of possible values.

Discrete variables arise from a counting process. Example: number of cars at a light, number of students in a classroom, number of rooms in a house.

- ✓ **Continuous (something that can be measured)** – has an infinite number of possible values that are not countable.

Continuous variables arise from a measuring process. Example: height, weight, age, miles per gallon, time.

- ➢ **Raw scores or data**: Numbers obtained in a particular situation. A collection of raw scores is usually called a distribution of scores. Data are individual facts or items of information.

Examples of a distribution:

- Test scores on an exam in a particular class.
- Ages of students at MCCC.
- IQ's of a random sample of 6^{th} grade students in the Trenton school district.

- ➢ **Population** – All people or things being considered in a particular situation.

A population consists of all the items or individuals or subjects about which you want to draw a conclusion. So, the population is the "large group" in which you are interested.

Example of the population:

- All students in the 6^{th} grade in the Trenton school district.
- The mean IQ score of all 6^{th} grade students in the Trenton school district.

- ✓ A **parameter** is a numerical summary of a population. Parameter is a numerical measure that describes a characteristic of a population.

Fundamental Issues of Statistics

➢ **Sample** – any portion (subset) of a population under consideration.

A sample is the portion of a population selected for analysis. The sample is the "small group" for whom we have (or plan to have) data, often randomly selected.

Examples of the sample:

- Fifty 6th grade students from the Trenton school district.
- The mean IQ score for fifty 6th grade students from the Trenton school district.

✓ A **statistic** is a numerical summary of a sample. Statistic is a numerical measure that describes a characteristic of a sample

✓ A **parameter** goes with a population and a **statistic** goes with a sample.

o **Random Sample**: A sample selected in such a way that every member of the population has an equal chance of being selected. The members of the random sample are picked arbitrarily from the population.

Explanation:
Population: All students attending Mercer County Community College Variable: Some measure of mathematical ability

Sample: Students leaving a section of calculus at MCCC.

This is not a random sample from the population of all students at MCCC. From this sample we should not attempt to infer anything about the mathematical ability of all students at MCCC.

A bias in obtaining a sample will destroy the value of the statistical information obtained since statistical inferences made from this information would be invalid.

Why use a sample instead of a population?

- Time
- Money
- Population continuously changing.
- Cannot use the entire population.

Primary & Secondary Data:
Primary data are the original data derived from your research endeavors. Secondary data are data derived from your primary data. Primary data is information collected through original or first-hand research. For example, surveys and focus group discussions. On the other hand, secondary data is information which has been collected in the past by someone else. For example, researching the internet, newspaper articles and company reports.

Basic Terms of Frequency Table

Let us consider the following frequency distribution table consisting the weights of 50 students.

Table: Frequency distribution table for the weight of 50 students				
Class Interval (Class)	Class Boundary (Original Class)	Class Mark (Mid Value) (x)	Frequency (f)	Cumulative Frequency (F)
54 - 57	53.5 - 57.5	55.5	6	6
58 - 61	57.5 - 61.5	59.5	9	15
62 - 65	61.5 - 65.5	63.5	11	26
66 - 69	65.5 - 69.5	67.5	16	42
70 - 73	69.5 - 73.5	71.5	8	50

A **frequency distribution** shows us a summarized grouping of data divided into mutually exclusive classes and the number of occurrences in a class. It is a way of showing unorganized data notably to show results of an event considered for a certain interest.

The **frequency distribution** table is an arrangement of the values that one or more variables take in a sample. Each entry in the table contains the frequency or count of the occurrences of values within a particular group or interval, and in this way, the table summarizes the distribution of values in the sample.

The **frequency distribution** is a representation, either in a graphical or tabular format that displays the number of observations within a given interval. The interval size depends on the data being analyzed and the goals of the analyst. The intervals must be mutually exclusive and exhaustive.

In statistics, a **frequency distribution** is a list, table or graph that displays the frequency of various outcomes in a sample. Each entry in the table contains the frequency or count of the occurrences of values within a particular group or interval.

The **class interval** (or **class width**) is the same for all classes. The classes all taken together must cover at least the distance from the lowest value (minimum) in the data to the highest (maximum) value. **Equal class** intervals are preferred in a frequency distribution, while **unequal class** intervals (for example logarithmic intervals) may be necessary for certain situations to produce a good spread of observations between the classes and avoid a large number of empty, or almost empty classes.

Corresponding to a class interval, the **class limits** may be defined as the minimum value and the maximum value the class interval may contain. The minimum value is known as the **lower-class limit (LCL)** and the maximum value is known as the **upper-class limit (UCL)**.

The **class boundaries** may be defined as the actual class limit of a class interval. For overlapping classification or mutually exclusive classification, the class boundaries coincide with the class limits. This is usually done for a continuous variable. However, for non-overlapping or mutually inclusive classification, we have **lower-class boundaries (LCB)** and **upper-class boundaries (UCB)** will have the following forms.

$$LCB = LCL - \frac{D}{2} \quad \& \quad UCB = UCL + \frac{D}{2}$$

where D is the difference between the LCL of the next class interval and the UCL of the given class interval.

The **class midpoint** (or **class-mark**) is a specific point in the center of the classes in a frequency distribution table. It’s also the center of a bar in a histogram. It is defined as the average of the upper and lower class limits. The lower-class limit is the lowest value in a class and the upper-class limits are the highest values that can be in the class. In other words, in a class interval, **class mid-point** may be defined as an arithmetic mean or average of the class limits or the class boundaries.

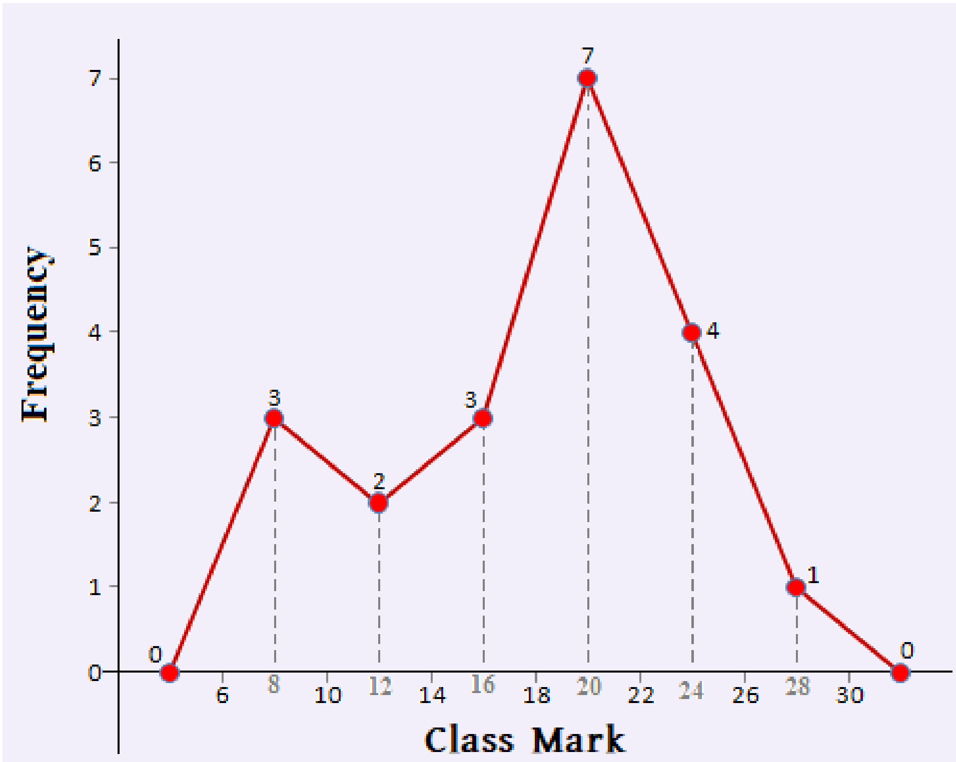
The **frequency** (or **absolute frequency**) of an event is the number of times the event occurred in an experiment or study. These frequencies are often graphically represented in histograms.

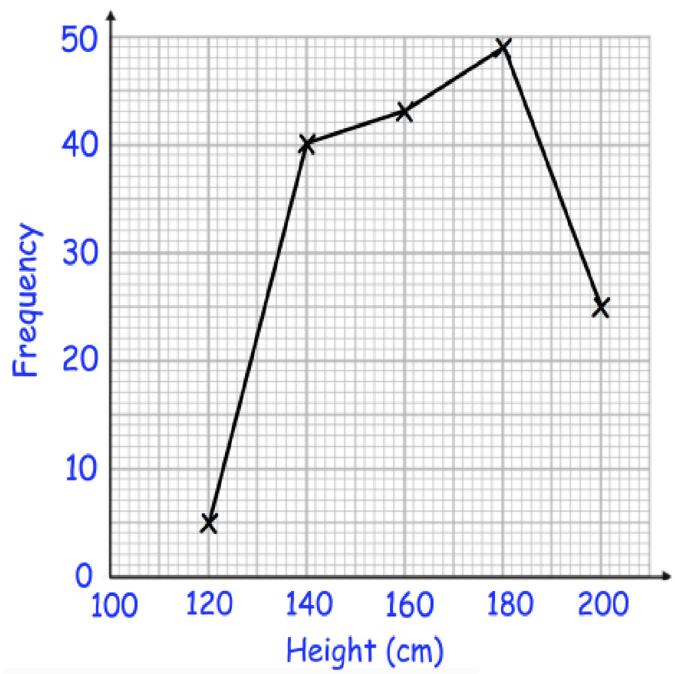
Cumulative frequency is defined as a running total of frequencies. The frequency of an element in a set refers to how many of that element there are in the set. Cumulative frequency can also be defined as the sum of all previous frequencies up to the current point.

Cumulative frequency analysis is the analysis of the frequency of occurrence of values of a phenomenon less than a reference value. The phenomenon may be time- or space-dependent. Cumulative frequency is also called the frequency of non-exceedance.

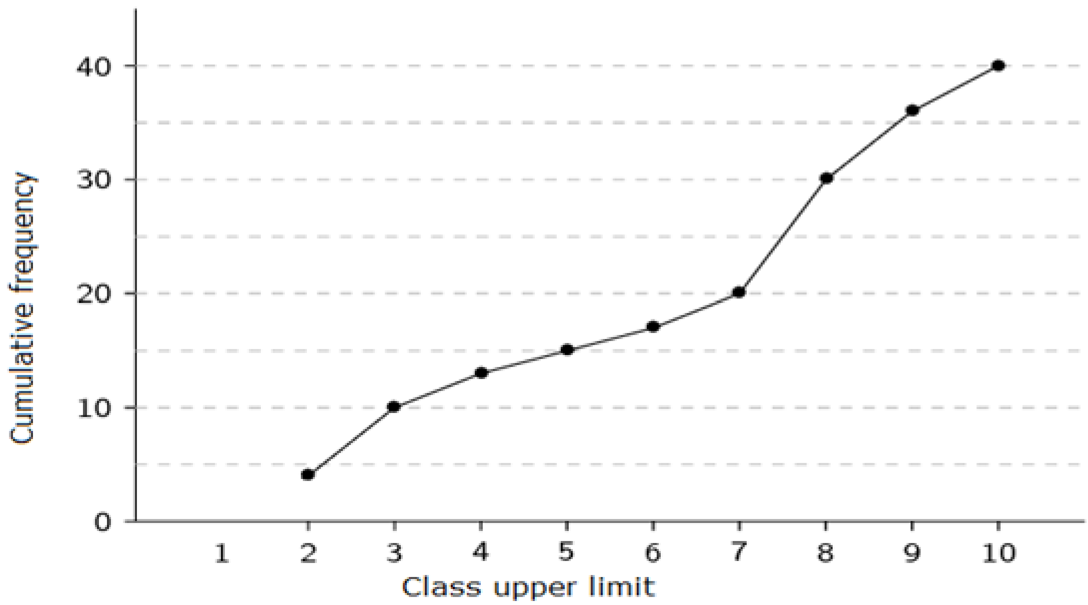
Statistical Graphs

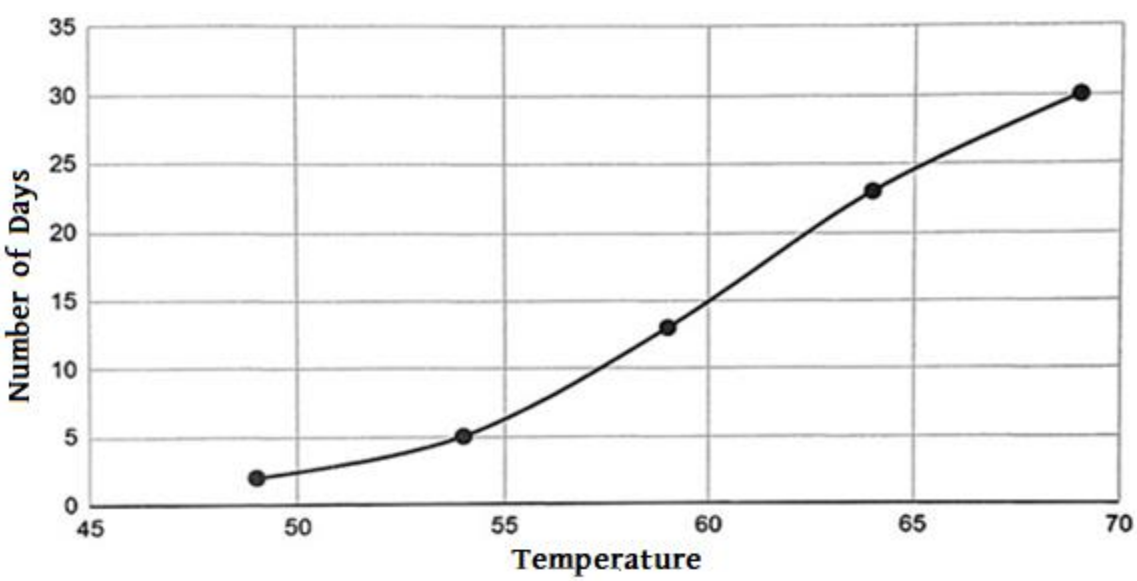
Frequency polygon: They are formed by lines. On the horizontal axis is the independent variable (class marks) and on the vertical axis is the dependent variable (frequency).





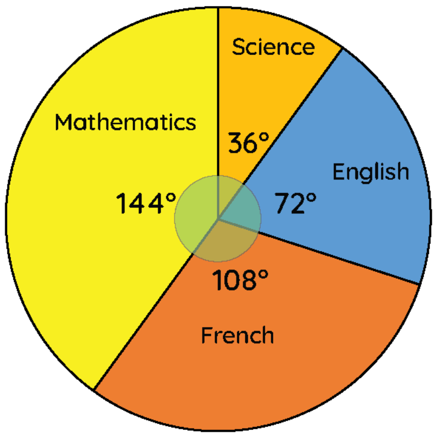
Cumulative frequency polygon: They are formed by increasing lines. On the horizontal axis is the independent variable (class upper limit) and on the vertical axis is the dependent variable (cumulative frequency).





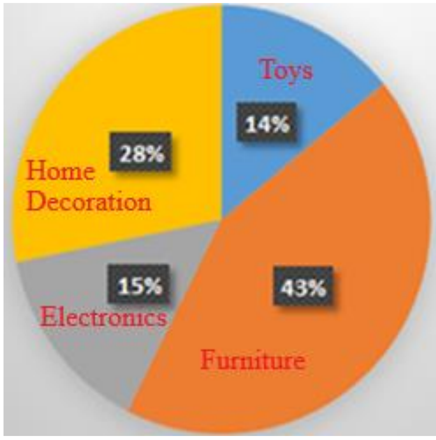
Pie chart: A circle is divided into sectors. The amplitude of each sector is proportional to the corresponding frequency.

$\theta_i = \frac{f_i}{\Sigma f_i} \times 360^o \rightarrow f_i = \frac{\theta_i \times \Sigma f_i}{360}$ and $p_i = \frac{f_i}{\Sigma f_i} \times 100\% \rightarrow f_i = \frac{p_i \times \Sigma f_i}{100}$



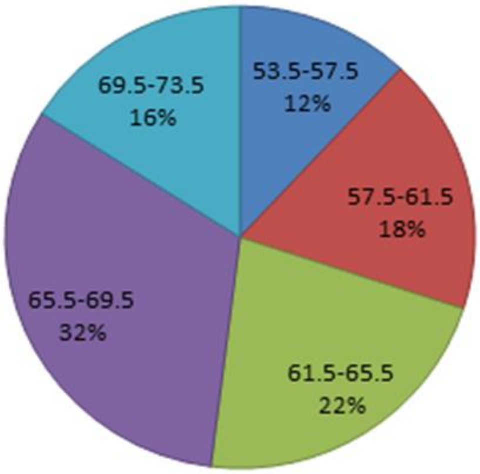
Consider the total number of students is 300.

Number secured	Alocated subject
$40 - 55$	French
$55 - 70$	English
$70 - 85$	Science
$85 - 100$	Mathematics

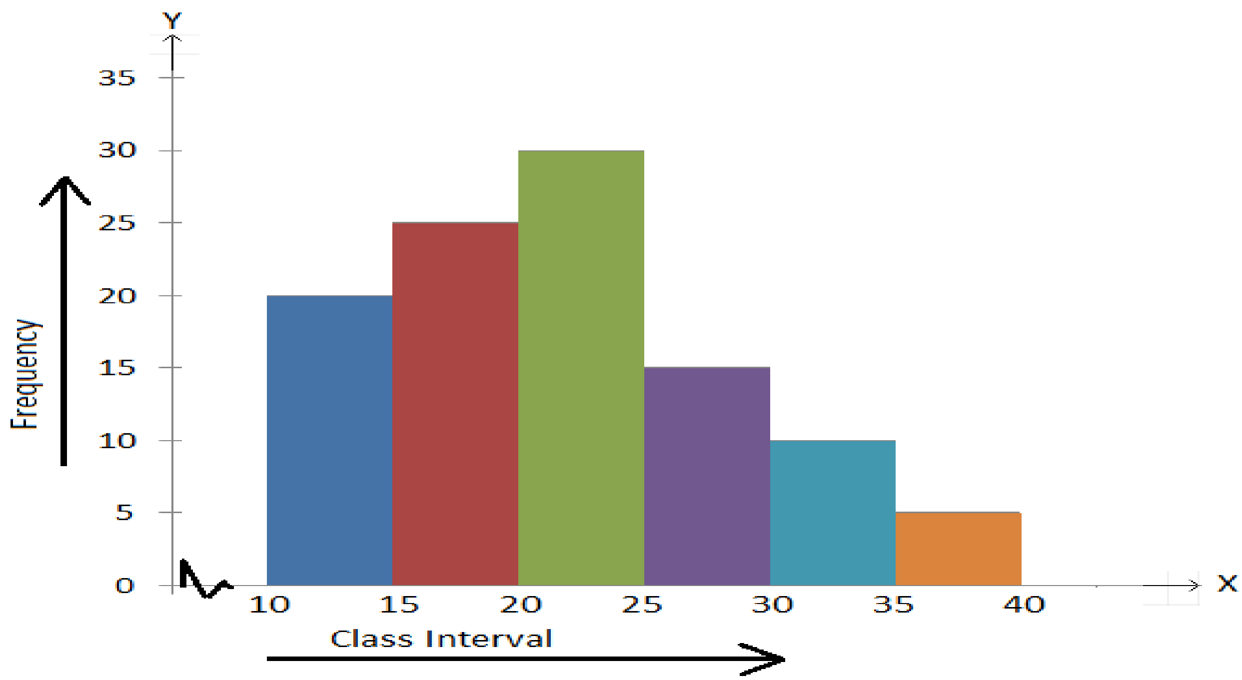


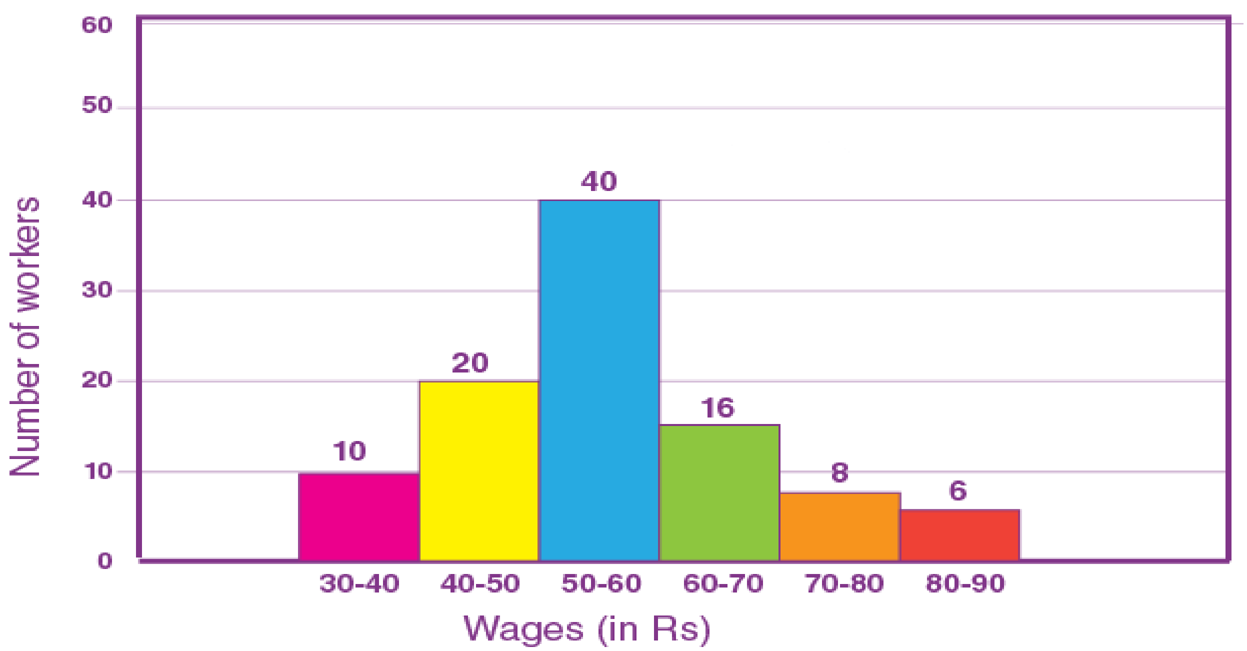
An women has to buy in total 150 domestic elements with the following price ranges and the percentage of the individual elements are as provided in the given pie-chart.

Toys $1000 - 2000$, furniture $3000 - 4000$, home decoration $4000 - 5000$, and electronics $2000 - 3000$.

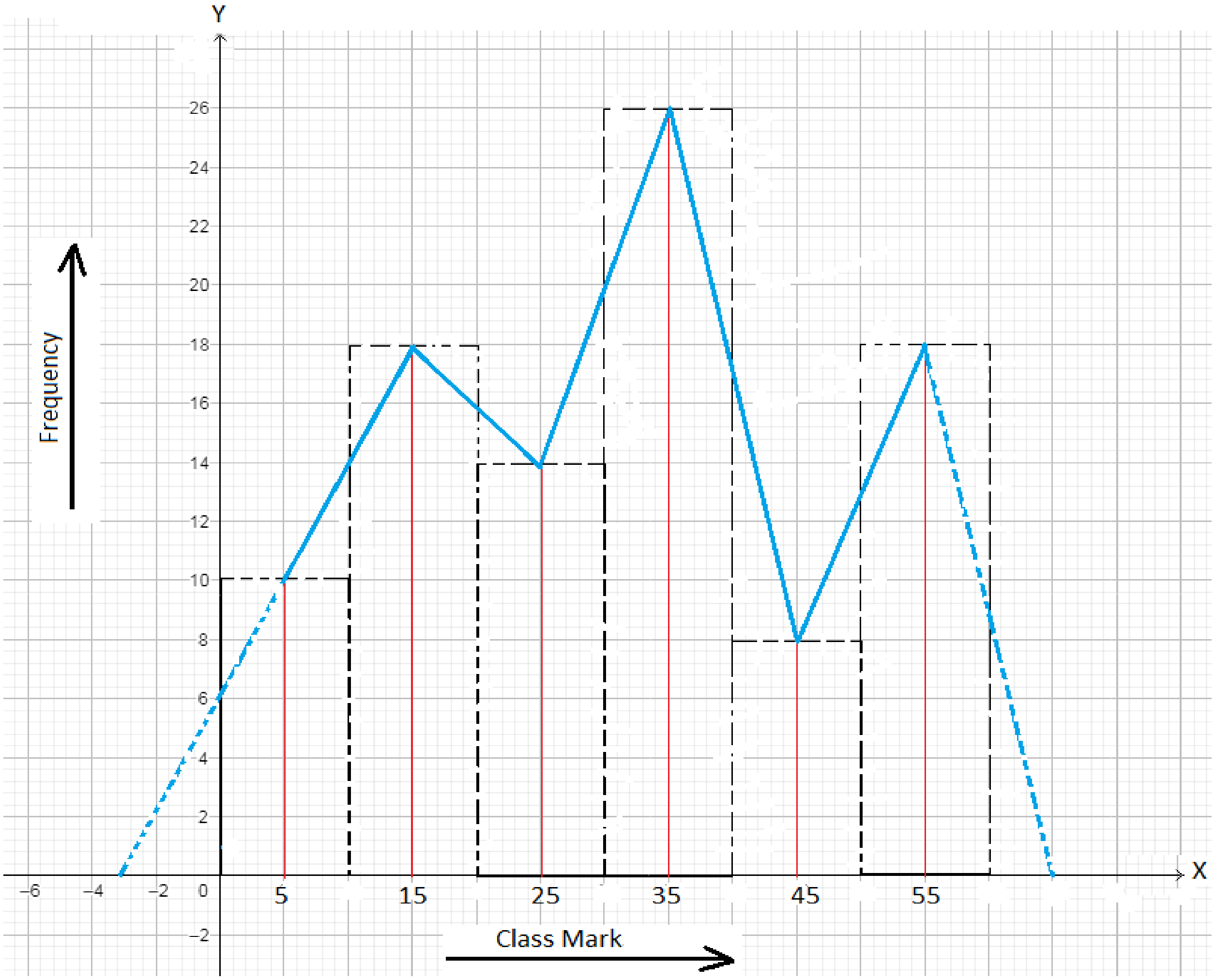


Histogram: It is a bar graph in which the height of these bars is proportional to the frequency. There is no space between bars. It is only used if the variable is quantitative and the scale of the values is continuous.





The relation between the frequency polygon and the histogram:



Please, visit the following links for more details.

https://www.statisticshowto.datasciencecentral.com/

https://www.tutorialspoint.com/statistics/index.htm

https://people.richland.edu/james/lecture/m170/ch02-def.html

https://www.scribbr.com/statistics/

https://libguides.library.curtin.edu.au/uniskills/numeracy-skills/statistics

https://www.statisticshowto.com/probability-and-statistics/

https://courses.lumenlearning.com/introstats1/chapter/learning-outcomes/

Measures of Central Tendency

Central tendency: A measure of central tendency is a single value that attempts to describe a set of data by identifying the central position within that set of data. As such, measures of central tendency are sometimes called measures of central location.

The Mean, Mode (Mo) and Median (Me) are all valid measures of central tendency, but under different conditions, some measures of central tendencies, such as Quartile, Decile, and Percentile become more appropriate to use than others.

There are three types of Mean, namely Arithmetic Mean (AM), Geometric Mean (GM), and Harmonic Mean (HM).

For n number of classes Arithmetic Mean can be estimated as

$$AM = \bar{x} = \frac{\sum_{i=1}^{n} f_i x_i}{\sum_{i=1}^{n} f_i}$$

For the shift a and scale h, the coding formula for Arithmetic Mean can be found as

$$u_i = \frac{x_i - a}{h}$$

$$\Rightarrow x_i = a + hu_i$$

$$\Rightarrow f_i x_i = af_i + hf_i u_i$$

$$\Rightarrow \sum_{i=1}^{n} f_i x_i = a \sum_{i=1}^{n} f_i + h \sum_{i=1}^{n} f_i u_i$$

$$\Rightarrow \frac{\sum_{i=1}^{n} f_i x_i}{\sum_{i=1}^{n} f_i} = a + h \frac{\sum_{i=1}^{n} f_i u_i}{\sum_{i=1}^{n} f_i}$$

$$\text{So, } AM = a + h \frac{\sum_{i=1}^{n} f_i u_i}{\sum_{i=1}^{n} f_i}$$

For n number of classes Geometric Mean can be estimated as

$$GM = \left(\prod_{i=1}^{n} {x_i}^{f_i} \right)^{\frac{1}{\sum_{i=1}^{n} f_i}}$$

For the feasible estimation, the working formula for Geometric Mean can be found as

$$\log(GM) = \log\left(\prod_{i=1}^{n} {x_i}^{f_i}\right)^{\frac{1}{\sum_{i=1}^{n} f_i}}$$

$$\Rightarrow \log(GM) = \frac{1}{\sum_{i=1}^{n} f_i} \log\left(\prod_{i=1}^{n} {x_i}^{f_i}\right)$$

$$\Rightarrow \log(GM) = \frac{\sum_{i=1}^{n} \log(x_i)^{f_i}}{\sum_{i=1}^{n} f_i}$$

$$\Rightarrow \log(GM) = \frac{\sum_{i=1}^{n} f_i \log(x_i)}{\sum_{i=1}^{n} f_i}$$

$$\text{So, } GM = \text{Antilog}\left(\frac{\sum_{i=1}^{n} f_i \log x_i}{\sum_{i=1}^{n} f_i}\right) = 10^{\left(\frac{\sum_{i=1}^{n} f_i \log x_i}{\sum_{i=1}^{n} f_i}\right)}$$

For n number of classes Harmonic Mean can be estimated as

$$HM = \left(\frac{\sum_{i=1}^{n} f_i \left(\frac{1}{x_i}\right)}{\sum_{i=1}^{n} f_i}\right)^{-1} = \left(\frac{\sum_{i=1}^{n} \left(\frac{f_i}{x_i}\right)}{\sum_{i=1}^{n} f_i}\right)^{-1} = \frac{\sum_{i=1}^{n} f_i}{\sum_{i=1}^{n} \left(\frac{f_i}{x_i}\right)}$$

Formula for Mode of the frequency distribution can be estimated as

$$Mo = L + \frac{\Delta_1}{\Delta_1 + \Delta_2} \times C$$

The class at which the highest frequency is present is called the modal class. L is the lower limit of the modal class, $\Delta_1 = f_m - f_{m-1}$ is the frequency difference between the modal and pre-modal class, $\Delta_2 = f_m - f_{m+1}$ is the frequency difference between the modal and post-modal class, and C is the class size.

Formula for Median of the frequency distribution can be estimated as

$$Me = L + \frac{\frac{N}{2} - F_{m-1}}{f_m} \times C$$

The class at which $\frac{N}{2} - th$ frequency ($N = \sum_{i=1}^{n} f_i$) is present is called the median class. L is the lower limit of the median class, F_{m-1} is the cumulative frequency pre-median class, f_m is the frequency of the median class, and C is the class size.

Formula for Quartile is $$Q_i = L + \frac{\frac{i\times N}{4} - F_{q-1}}{f_q} \times C\ ;\ i = 1,2,3$$

The class at which $\frac{i\times N}{4} - th$ frequency ($N = \sum_{i=1}^{n} f_i$) is present is called the $i - th$ quartile class. L is the lower limit of the quartile class, F_{q-1} is the cumulative frequency pre-quartile class, f_q is the frequency of the quartile class, and C is the class size.

Formula for Decile is $$D_i = L + \frac{\frac{i\times N}{10} - F_{d-1}}{f_d} \times C\ ;\ i = 1,2, \dots \dots, 9$$

The class at which $\frac{i\times N}{10} - th$ frequency ($N = \sum_{i=1}^{n} f_i$) is present is called the $i - th$ decile class. L is the lower limit of the decile class, F_{d-1} is the cumulative frequency pre-decile class, f_d is the frequency of the decile class, and C is the class size.

Formula for Prcentile is $$P_i = L + \frac{\frac{i\times N}{100} - F_{p-1}}{f_p} \times C\ ;\ i = 1,2, \dots \dots, 99$$

The class at which $\frac{i\times N}{100} - th$ frequency ($N = \sum_{i=1}^{n} f_i$) is present is called the $i - th$ percentile class. L is the lower limit of the percentile class, F_{p-1} is the cumulative frequency pre-percentile class, f_p is the frequency of the percentile class, and C is the class size.

$$AM - Mo = 3(AM - Me)$$

$$Mo = 3Me - 2Me$$

Measures of Dispersion

Dispersion: Dispersion in statistics is a way of describing how to spread out a set of data is. When a data set has a large value, the values in the set are widely scattered; when it is small the items in the set are tightly clustered.

The spread of a data set can be described by a range of descriptive statistics including Mean Deviation (MD), Standard Deviation (SD), and Interquartile Range. Those are called the absolute measures of dispersion.

Also, there are some relative measures of dispersion, such as co-efficient of Mean Deviation, co-efficient of Standard Deviation, and co-efficient of Interquartile Range.

There are three types of Mean Deviation, estimated from Arithmetic Mean, Mode, and Median, respectively.

$$MD_{AM} = \frac{\sum_{i=1}^{n} f_i |x_i - AM|}{\sum_{i=1}^{n} f_i}$$

$$MD_{Mo} = \frac{\sum_{i=1}^{n} f_i |x_i - Mo|}{\sum_{i=1}^{n} f_i}$$

$$MD_{Me} = \frac{\sum_{i=1}^{n} f_i |x_i - Me|}{\sum_{i=1}^{n} f_i}$$

Co-efficient of Mean Deviation is

$$CMD = \frac{MD_{Base}}{Base} \times 100\% \; ; \; Base = AM, Mo, Me$$

Variance and Standard Deviation of the statistical data are

$$V(X) = \sigma^2 = \frac{\sum_{i=1}^{n} f_i (x_i - AM)^2}{\sum_{i=1}^{n} f_i} = \frac{\sum_{i=1}^{n} f_i (x_i - \bar{x})^2}{\sum_{i=1}^{n} f_i}$$

$$SD = \sigma = \sqrt{\frac{\sum_{i=1}^{n} f_i (x_i - AM)^2}{\sum_{i=1}^{n} f_i}} = \sqrt{\frac{\sum_{i=1}^{n} f_i (x_i - \bar{x})^2}{\sum_{i=1}^{n} f_i}}$$

The working formula for Standard Deviation is

$$\sigma = \sqrt{\frac{\sum_{i=1}^{n} f_i (x_i - \bar{x})^2}{\sum_{i=1}^{n} f_i}} = \sqrt{\frac{\sum_{i=1}^{n} f_i (x_i^2 - 2x_i\bar{x} + \bar{x}^2)}{\sum_{i=1}^{n} f_i}} = \sqrt{\frac{\sum_{i=1}^{n} f_i x_i^2 - 2\bar{x}\sum_{i=1}^{n} f_i x_i + \bar{x}^2 \sum_{i=1}^{n} f_i}{\sum_{i=1}^{n} f_i}}$$

$$= \sqrt{\frac{\sum_{i=1}^{n} f_i x_i^2}{\sum_{i=1}^{n} f_i} - 2\bar{x}\frac{\sum_{i=1}^{n} f_i x_i}{\sum_{i=1}^{n} f_i} + \bar{x}^2} = \sqrt{\frac{\sum_{i=1}^{n} f_i x_i^2}{\sum_{i=1}^{n} f_i} - 2\frac{\sum_{i=1}^{n} f_i x_i}{\sum_{i=1}^{n} f_i}\frac{\sum_{i=1}^{n} f_i x_i}{\sum_{i=1}^{n} f_i} + \left(\frac{\sum_{i=1}^{n} f_i x_i}{\sum_{i=1}^{n} f_i}\right)^2}$$

$$= \sqrt{\frac{\sum_{i=1}^{n} f_i x_i^2}{\sum_{i=1}^{n} f_i} - 2\left(\frac{\sum_{i=1}^{n} f_i x_i}{\sum_{i=1}^{n} f_i}\right)^2 + \left(\frac{\sum_{i=1}^{n} f_i x_i}{\sum_{i=1}^{n} f_i}\right)^2} = \sqrt{\frac{\sum_{i=1}^{n} f_i x_i^2}{\sum_{i=1}^{n} f_i} - \left(\frac{\sum_{i=1}^{n} f_i x_i}{\sum_{i=1}^{n} f_i}\right)^2}$$

The coding formula for Standard Deviation is derived from $u_i = \frac{x_i - a}{h}$ as

$$x_i = a + hu_i$$

$$\Rightarrow \frac{\sum_{i=1}^{n} f_i x_i}{\sum_{i=1}^{n} f_i} = a + h\frac{\sum_{i=1}^{n} f_i u_i}{\sum_{i=1}^{n} f_i}$$

$$\Rightarrow \bar{x} = a + h\bar{x}$$

So, $x_i - \bar{x} = h(u_i - \bar{u})$

Then, we can write

$$\sigma = \sqrt{\frac{\sum_{i=1}^{n} f_i (x_i - \bar{x})^2}{\sum_{i=1}^{n} f_i}} = \sqrt{\frac{\sum_{i=1}^{n} f_i h^2 (u_i - \bar{u})^2}{\sum_{i=1}^{n} f_i}} = h \times \sqrt{\frac{\sum_{i=1}^{n} f_i (u_i - \bar{u})^2}{\sum_{i=1}^{n} f_i}}$$

Now, the working formula for Standard Deviation is

$$SD = h \times \sqrt{\frac{\sum_{i=1}^{n} f_i {u_i}^2}{\sum_{i=1}^{n} f_i} - \left(\frac{\sum_{i=1}^{n} f_i u_i}{\sum_{i=1}^{n} f_i}\right)^2}$$

Co-efficient of Variation is

$$CV = \frac{SD}{AM} \times 100\%$$

Interquartile Range is $IQR = Q_3 - Q_1$

Co-efficient of Interquartile Range $CIQR = \frac{Q_3 - Q_1}{Q_3 + Q_1} \times 100\%$

$$MD = \frac{4}{5} SD$$

$$IQR = \frac{2}{3} SD$$

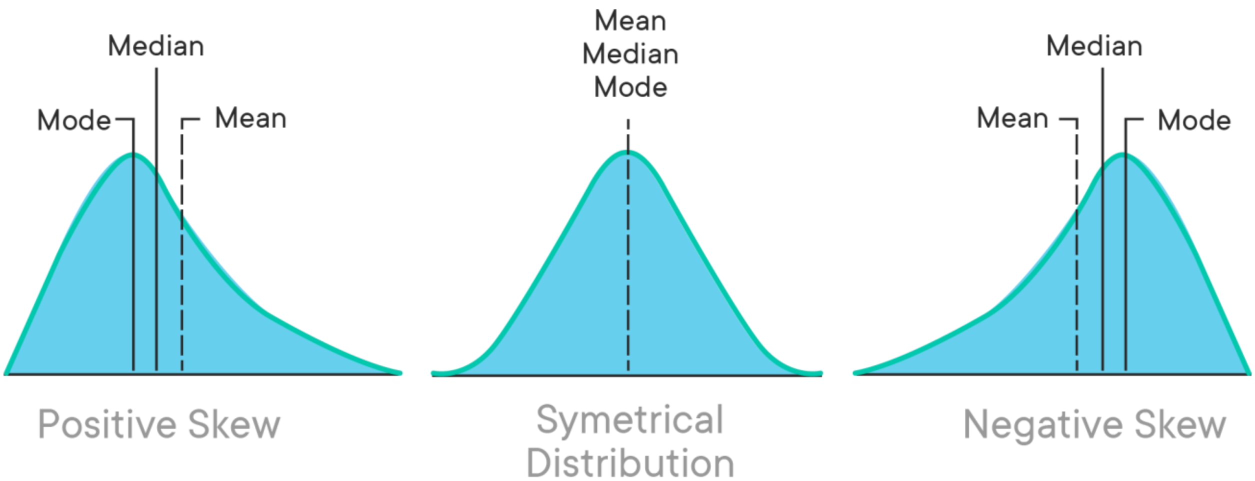
Statistical Index (z −score) of any given input

$$z_0 = \frac{x_0 - \bar{x}}{\sigma}$$

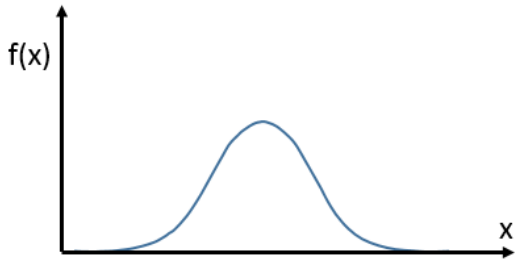
Moments, Skewness, and Kurtosis

Two distributions may have the same Mean and Standard Deviation but may differ in their shape of the distribution. Further description of their characteristics is necessary that is provided by measures of skewness and kurtosis. Moments are popularly used to describe the characteristics of a distribution. They represent a convenient and unifying method for summarizing many of the most commonly used descriptive statistical measures such as central tendency, variation, Skewness, and Kurtosis.

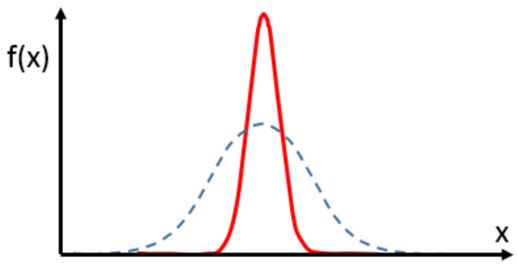
The term ‘skewness’ refers to a lack of symmetry or departure from symmetry, e.g., when a distribution is not symmetrical (or is asymmetrical) it is called a skewed distribution. The measures of skewness indicate the difference between the manners in which the observations are distributed in a particular distribution compared with the symmetrical (or normal) distribution. The concept of skewness gains importance from the fact that statistical theory is often based upon the assumption of the normal distribution. A measure of skewness is, therefore, necessary in order to guard against the consequence of this assumption.



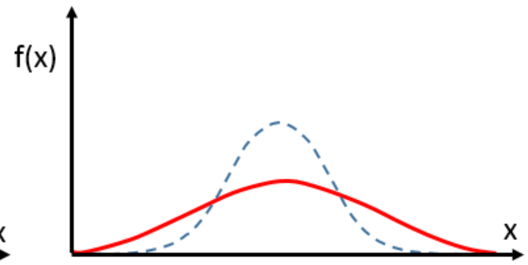
In statistics, kurtosis refers to the degree of flatness or peakedness in the region about the mode of a frequency curve. The degree of kurtosis of a distribution is measured relative to the flatness or peakedness of a normal curve, it is called “Platykurtic” or “Leptokurtic”. The normal curve itself is known as “Mesokurtic”.



Zero kurtosis
Normal distribution
(Mesokurtic)



Positive kurtosis
(Leptokurtic)



Negative kurtosis
(Platykurtic)

The $r-th$ raw moment about an arbitrary point A is

$$m_r' = \frac{\sum_{i=1}^{n} f_i (x_i - A)^r}{\sum_{i=1}^{n} f_i} \; ; A \neq \bar{x}$$

The coding formula for the $r-th$ raw moment from $u_i = \frac{x_i - A}{h}$ is

$$x_i - A = hu_i$$

Then, we can write

$$m_r' = \frac{\sum_{i=1}^{n} f_i h^r u_i^r}{\sum_{i=1}^{n} f_i} = h^r \times \frac{\sum_{i=1}^{n} f_i u_i^r}{\sum_{i=1}^{n} f_i}$$

The $r-th$ central moment about the arithmetic mean $\bar{x}$ is

$$m_r = \frac{\sum_{i=1}^{n} f_i (x_i - \bar{x})^r}{\sum_{i=1}^{n} f_i}$$

We can write

$$m_1 = \frac{\sum_{i=1}^{n} f_i (x_i - \bar{x})}{\sum_{i=1}^{n} f_i} = \frac{\sum_{i=1}^{n} f_i x_i - \bar{x} \sum_{i=1}^{n} f_i}{\sum_{i=1}^{n} f_i} = \frac{\sum_{i=1}^{n} f_i x_i}{\sum_{i=1}^{n} f_i} - \bar{x} = \bar{x} - \bar{x} = 0$$

So, $m_1 = 0$ for every dataset.

Again. we can write

$$m_2 = \frac{\sum_{i=1}^{n} f_i (x_i - \bar{x})^2}{\sum_{i=1}^{n} f_i} = \sigma^2$$

So, m_2 is the variance of the data set and hence $SD = \sigma = \sqrt{m_2}$.

Estimation of the central moments from the raw moments

$$m_1 = 0$$

$$m_2 = m_2' - {m_1'}^2$$

$$m_3 = m_3' - 3m_2'm_1' + 2{m_1'}^3$$

$$m_4 = m_4' - 4m_3'm_1' + 6m_2'{m_1'}^2 - 3{m_1'}^4$$

Co-efficient of Skewness

$$\gamma_3 = \frac{m_3}{\sqrt{m_2^3}}$$

If $\gamma_3 < 0$, the provided data set is called negatively skewed.

If $\gamma_3 = 0$, the provided data set is called non-skewed (Normal).

If $\gamma_3 > 0$, the provided data set is called positively skewed.

Co-efficient of Kurtosis

$$\gamma_4 = \frac{m_4}{m_2^2}$$

If $\gamma_4 < 3$, the provided data set is called platykurtic (flattered).

If $\gamma_4 = 3$, the provided data set is called mesokurtic (balanced).

If $\gamma_4 > 3$, the provided data set is called leptokurtic (peaked).

Corrected central momnets due to class size c as chosen as the round figure

$$m_1^{(corrected)} = 0$$

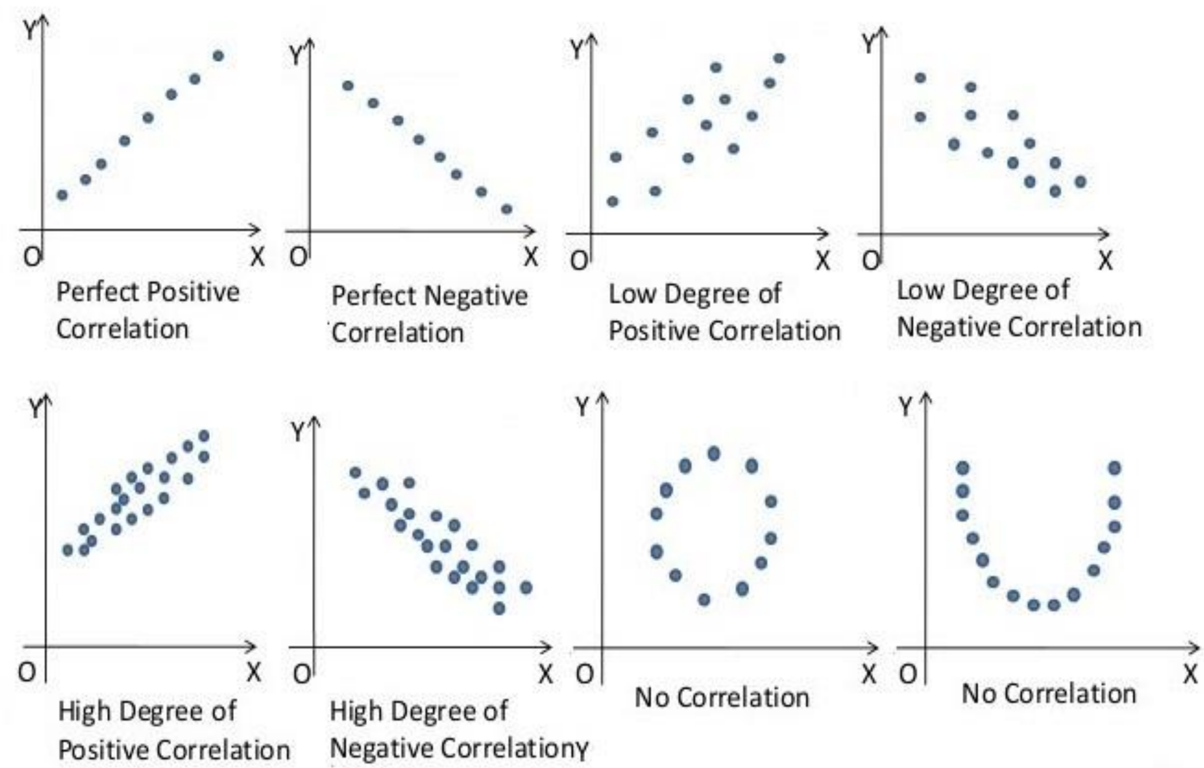
$$m_2^{(corrected)} = m_2 - \frac{1}{12}c^2$$

$$m_3^{(corrected)} = m_3$$

$$m_4^{(corrected)} = m_4 - \frac{1}{2}m_2c^2 - \frac{7}{240}c^4$$

Correlation and Regression

Correlation Analysis: Correlation analysis is applied in quantifying the association between two continuous variables, for example, a dependent and independent variable or among two independent variables.

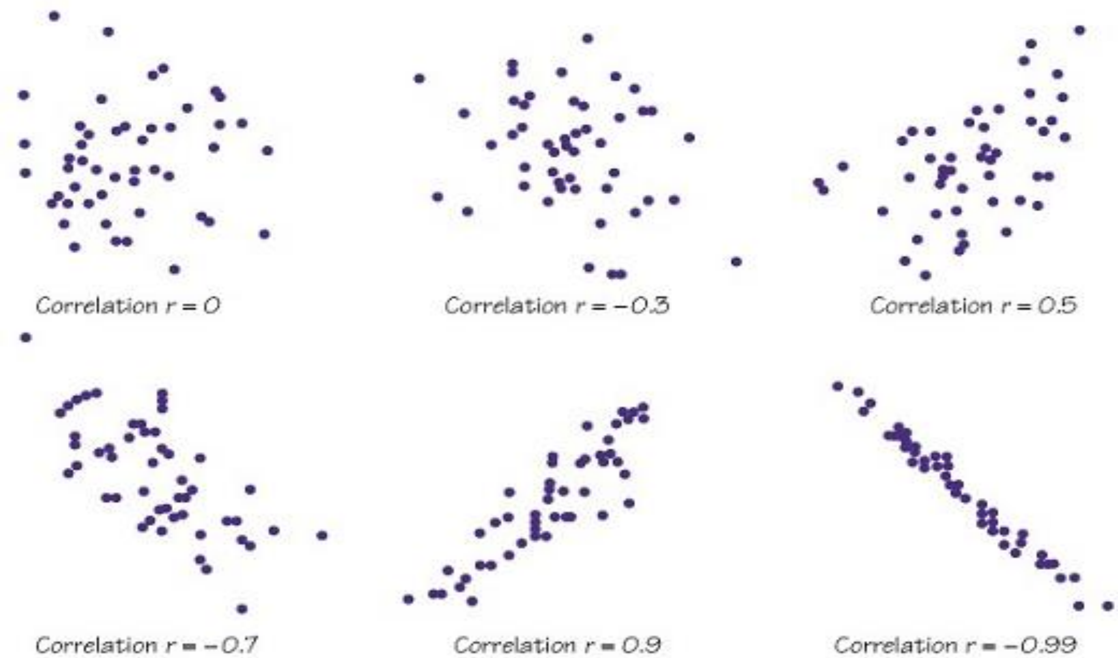


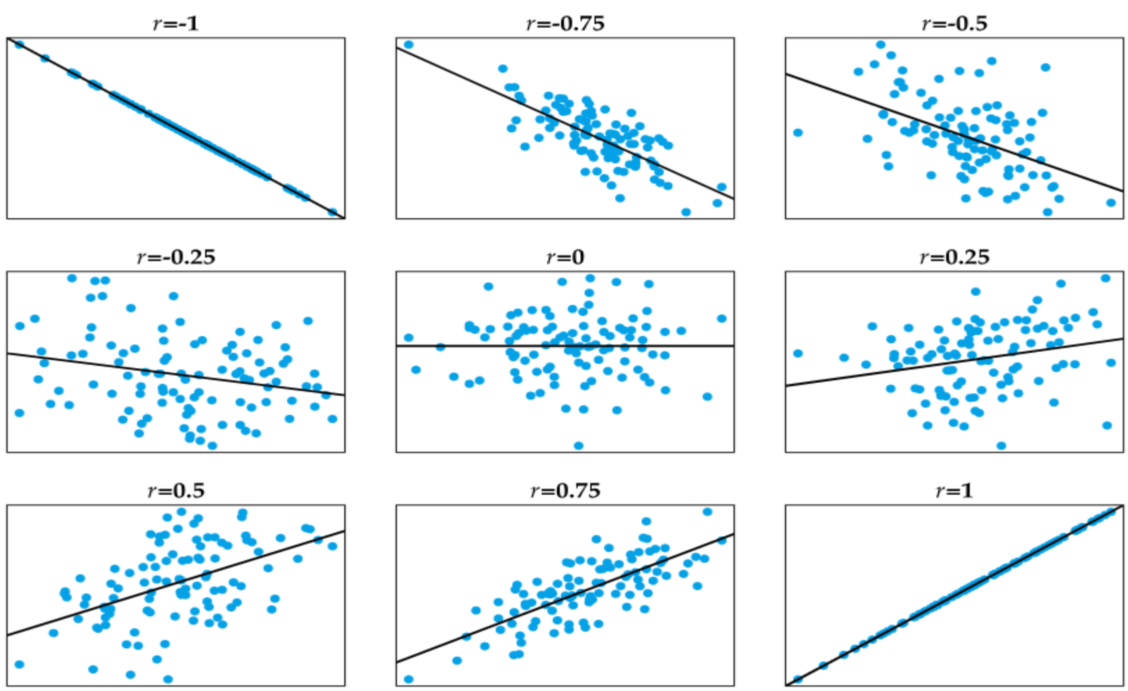
Scatter diagram

The sign of the coefficient of correlation is denoted by r that ranges between -1 and $+1$, and shows the direction of the association. The magnitude of the coefficient shows the strength of the association. The sample of a correlation coefficient is estimated in the correlation analysis.

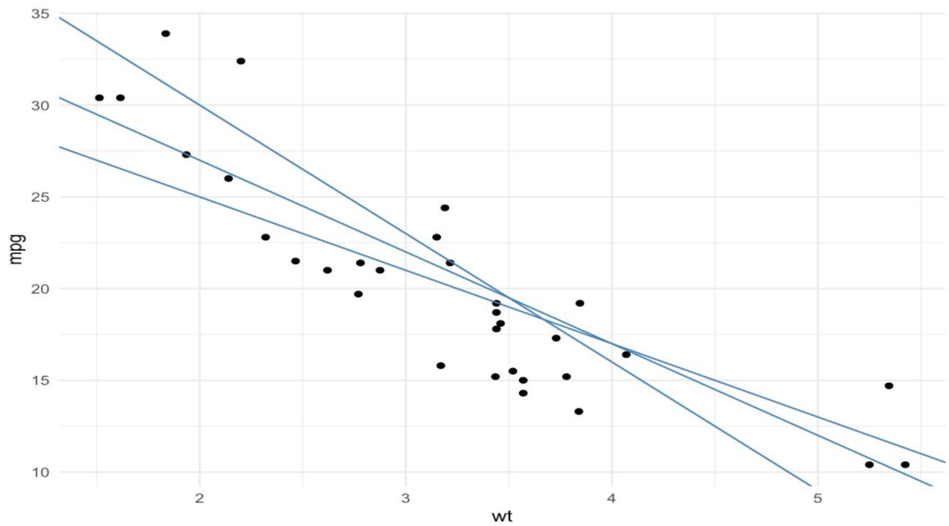
Value of r	Strength
$r = 0$	No correlation
$-0.5 < r < 0$ or $0 < r < 0.5$	Weak/low-degree correlation
$-1 < r < -0.5$ or $0.5 < r < 1$	Strong/high-degree correlation
$r = \pm 1$	Strict/perfect correlation

Correlation coefficient quantifies the strength and direction of the linear association among two variables. The correlation among two variables can either be positive, i.e., a higher level of one variable is related to a higher level of another or negative, i.e., a higher level of one variable is related to a lower level of the other.

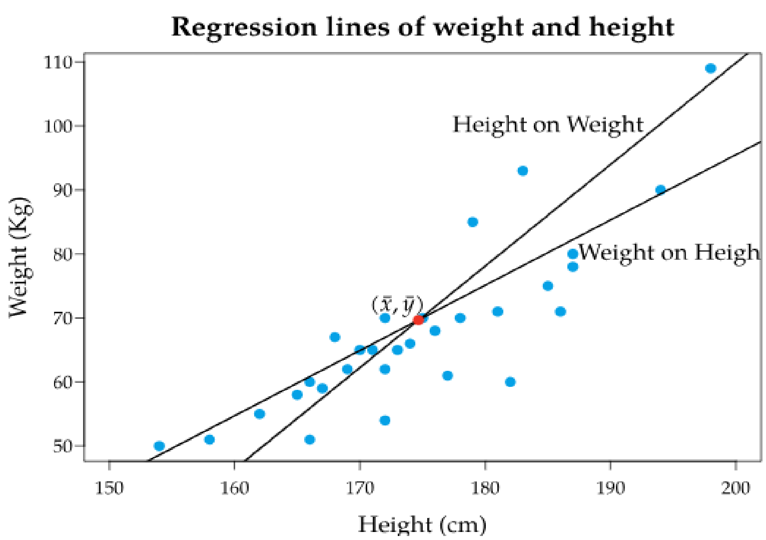




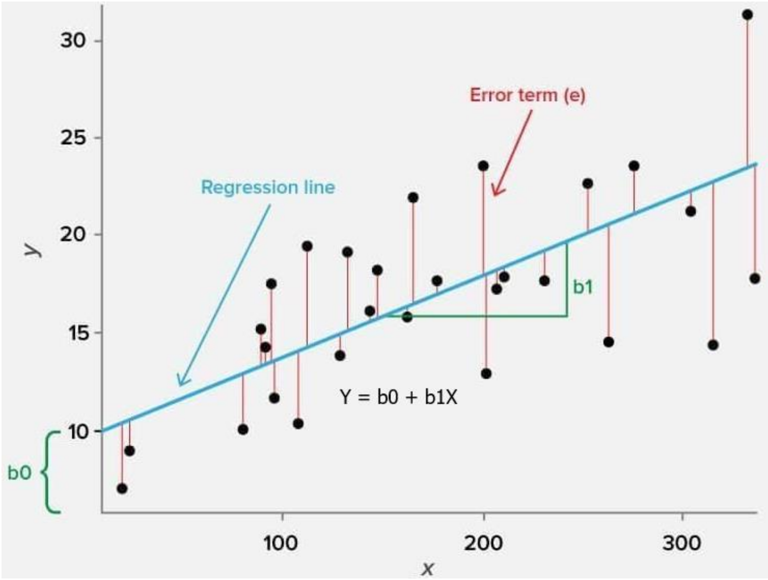
Regression Analysis: Regression analysis involves identifying the relationship between a dependent variable and one or more variables. The outcome variable is known as the dependent or response variable and the risk elements, and cofounders are known as predictors or independent variables. The dependent variable is shown by $\boldsymbol{y}$ and independent variables are shown by $\boldsymbol{x}$ in regression analysis.



A model of the relationship is hypothesized, and estimates of the parameter values are used to develop an estimated regression equation. Various tests are then employed to determine if the model is satisfactory. If the model is deemed satisfactory, the estimated regression equation can be used to predict the value of the dependent variable given values for the independent variables.



Linear Regression: This is a linear approach to modeling the relationship between the scalar components and one or more independent variables. If the regression has one independent variable, then it is known as a simple linear regression. If it has more than one independent variable, then it is known as multiple linear regression.



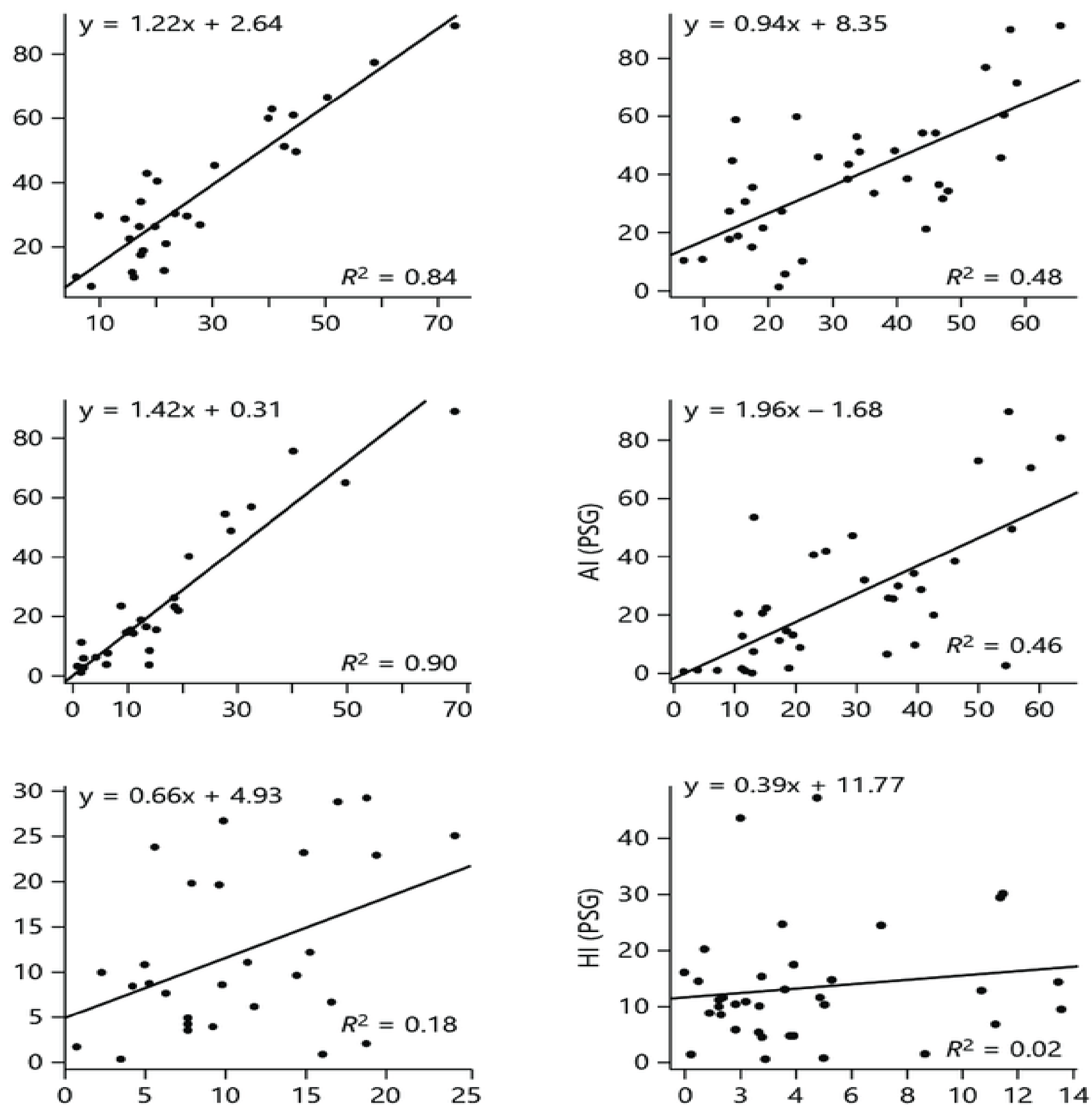
Linear regression only focuses on the conditional probability distribution of the given values rather than the joint probability distribution. In general, all the real-world regression models involve multiple predictors. So, the term linear regression often describes multivariate linear regression.

Coefficient of Determination: The term r^2 is known as the coefficient of determination, with a range between 0 and 1, which is used as a key statistical measure in regression analysis. It indicates how well the independent variable(s) explain the variation in the dependent variable.

The coefficient of determination is a vital tool for model comparison, allowing analysts to quickly assess which of several regression models best explains the target variable. It also serves to quantify the explanatory strength of a relationship in research, providing a solid measure for how much observed variation is captured by the model. Moreover, it acts as a key communication tool, translating model fit into an intuitive percentage of variance explained for reports and stakeholder discussions.

Assume a model is built to predict a student's exam score (Y) based on hours studied (X). A value of r^2 is attained 0.80. So, the model explains 80% of the variation in exam scores among the students in your sample. The remaining 20% of the variation is due to other factors of the model doesn't capture, such as natural ability, sleep, and question luck.

Sometimes r^2 can be misleading because it always increases when more variables are added, even irrelevant ones, which can lead to overfitting. It does not imply causation and cannot determine if the model's assumptions are violated. Importantly, r^2 value is highly field-dependent as well, so a high score doesn't guarantee an accurate or useful predictive model.



Comparison between Correlation and Regression:

Topic	Correlation	Regression
Meaning	Correlation is a statistical measure that determines a co-relationship or association of two variables.	Regression describes how an independent variable is numerically related to a dependent variable.
Main purpose	Correlation analysis lets experimenters know the association or the absence of a relationship between two variables under study; if the variables are correlated, it allows measuring the strength of their association.	Regression analysis helps determine a functional relationship between two variables so as to estimate the unknown variable with the help of known variable(s) and make future projections on events.
Objective	To find a numerical value that expresses the relationship between the variables.	To estimate the values of a random variable based on the values of a fixed variable.
Usage	Represents the linear relationship between two variables.	Fits the best line and estimates one variable based on another variable.
Nature of variables	The variables are not designated as dependent or independent.	One variable is dependent, and another variable is independent.
Indicates	The correlation coefficient indicates the extent to which two variables move together.	Regression indicates the impact of a unit change in the known variable on the estimated variable.
Range	Correlation coefficients can range from -1 to $+1$.	In regression analysis, if the magnitude of a regression coefficient is more than 1, then the magnitude of the other regression coefficients is less than 1.
Nature of the coefficient	The correlation coefficient is symmetric and mutual.	The regression coefficient is not symmetric.
Exceptional cases	Non-sense correlation may exist in the correlation analysis.	Non-sense regression doesn’t exist in regression analysis.
Association	The correlation coefficient measures the extent and direction of a linear association between two variables.	Linear regression describes one variable as a linear function of another variable.
Relationship	Correlation is confined to the linear relationship between variables only.	Regression studies linear and non-linear relationships.
Scope	Correlation analysis has limited applications.	Regression analysis has wider applications.

Fundamental Issues of Statistics

For N is the number of inputs, and x & y are two variables, there are some well-known notations

$$V(X) = \sigma_x^2 = \frac{\sum(x-\bar{x})^2}{N} = S_{xx}$$

$$V(Y) = \sigma_y^2 = \frac{\sum(y-\bar{y})^2}{N} = S_{yy}$$

$$Cov(X,Y) = \sigma_{xy} = \frac{\sum(x-\bar{x})(y-\bar{y})}{N} = S_{xy}$$

Here, S_{xy} is called the covariance between x & y.

The correlation coefficient

$$r_{xy} = r_{yx} = \frac{\sigma_{xy}}{\sigma_x\sigma_y} = \frac{\frac{\sum(x-\bar{x})(y-\bar{y})}{N}}{\sqrt{\frac{\sum(x-\bar{x})^2}{N}\frac{\sum(y-\bar{y})^2}{N}}} = \frac{\sum(x-\bar{x})(y-\bar{y})}{\sqrt{\sum(x-\bar{x})^2\sum(y-\bar{y})^2}}$$
$$= \frac{N\sum xy - \sum x \sum y}{\sqrt{(N\sum x^2 - (\sum x)^2)(N\sum y^2 - (\sum y)^2)}}$$

There are two Regression co-efficients

$$b_{\frac{y}{x}} = \frac{\sigma_{xy}}{\sigma_x^2} = \frac{\frac{\sum(x-\bar{x})(y-\bar{y})}{N}}{\frac{\sum(x-\bar{x})^2}{N}} = \frac{\sum(x-\bar{x})(y-\bar{y})}{\sum(x-\bar{x})^2} = \frac{N\sum xy - \sum x\sum y}{N\sum x^2 - (\sum x)^2}$$

$$b_{\frac{x}{y}} = \frac{\sigma_{xy}}{\sigma_y^2} = \frac{\frac{\sum(x-\bar{x})(y-\bar{y})}{N}}{\frac{\sum(y-\bar{y})^2}{N}} = \frac{\sum(x-\bar{x})(y-\bar{y})}{\sum(y-\bar{y})^2} = \frac{N\sum xy - \sum x\sum y}{N\sum y^2 - (\sum y)^2}$$

The Regression lines can be formed as given below.

Regression line of y on x is

$$\hat{y} - \bar{y} = b_{\frac{y}{x}}(x-\bar{x})$$
$$\Rightarrow \hat{y} - \frac{\sum y}{N} = b_{\frac{y}{x}}\left(x - \frac{\sum x}{N}\right)$$
$$\Rightarrow \hat{y} = b_{\frac{y}{x}}\left(x - \frac{\sum x}{N}\right) + \frac{\sum y}{N}$$
$$\Rightarrow \hat{y} = b_{\frac{y}{x}}(x) + \left(\frac{\sum y}{N} - b_{\frac{y}{x}}\frac{\sum x}{N}\right)$$

Regression line of x on y is

$$\hat{x} - \bar{x} = b_{\frac{x}{y}}(y-\bar{y})$$
$$\Rightarrow \hat{x} - \frac{\sum x}{N} = b_{\frac{x}{y}}\left(y - \frac{\sum y}{N}\right)$$
$$\Rightarrow \hat{x} = b_{\frac{x}{y}}\left(y - \frac{\sum y}{N}\right) + \frac{\sum x}{N}$$
$$\Rightarrow \hat{x} = b_{\frac{x}{y}}(y) + \left(\frac{\sum x}{N} - b_{\frac{x}{y}}\frac{\sum y}{N}\right)$$

Assume $u = \frac{x-a}{h}$ & $v = \frac{y-b}{k}$, where a & b are the shifts and h & k are the scales. Then it can be shown that

$$x = a + hu$$

That gives, $\bar{x} = a + h\bar{u}$

So, $x - \bar{x} = h(u - \bar{u})$

$$y = b + kv$$

That gives, $\bar{y} = b + k\bar{v}$

So, $y - \bar{y} = k(v - \bar{v})$

Then the correlation coefficient

$$r_{xy} = \frac{\sum(x-\bar{x})(y-\bar{y})}{\sqrt{\sum(x-\bar{x})^2 \sum(y-\bar{y})^2}} = \frac{hk\sum(u-\bar{u})(v-\bar{v})}{\sqrt{h^2k^2\sum(u-\bar{u})^2\sum(v-\bar{v})^2}} = \frac{\sum(u-\bar{u})(v-\bar{v})}{\sqrt{\sum(u-\bar{u})^2\sum(v-\bar{v})^2}} = r_{uv}$$
$$= \frac{N\sum uv - \sum u \sum v}{\sqrt{(N\sum u^2 - (\sum u)^2)(N\sum v^2 - (\sum v)^2)}}$$

So, the Correlation coefficient is independent of the shift and scale.

Again, the Regression co-efficients

$$\boldsymbol{b_{\frac{y}{x}} = \frac{\sum(x-\bar{x})(y-\bar{y})}{\sum(x-\bar{x})^2} = \frac{hk\sum(u-\bar{u})(v-\bar{v})}{h^2\sum(u-\bar{u})^2} = \frac{k\sum(u-\bar{u})(v-\bar{v})}{h\sum(u-\bar{u})^2} = \frac{k}{h}\times b_{\frac{v}{u}}}$$
$$\boldsymbol{= \frac{k}{h}\times\left(\frac{N\sum uv - \sum u\sum v}{N\sum u^2 - (\sum u)^2}\right)}$$

$$b_{\frac{x}{y}} = \frac{\sum(x-\bar{x})(y-\bar{y})}{\sum(y-\bar{y})^2} = \frac{hk\sum(u-\bar{u})(v-\bar{v})}{k^2\sum(v-\bar{v})^2} = \frac{h\sum(u-\bar{u})(v-\bar{v})}{k\sum(v-\bar{v})^2} = \frac{h}{k}\times b_{\frac{u}{v}}$$
$$= \frac{h}{k}\times\left(\frac{N\sum uv - \sum u\sum v}{N\sum v^2 - (\sum v)^2}\right)$$

So, the Regression coefficients are independent of the shifts but dependent on scales.

Now, we have

$$b_{\frac{y}{x}} \times b_{\frac{x}{y}} = \frac{\sum(x-\bar{x})(y-\bar{y})}{\sum(x-\bar{x})^2} \times \frac{\sum(x-\bar{x})(y-\bar{y})}{\sum(y-\bar{y})^2} = \frac{(\sum(x-\bar{x})(y-\bar{y}))^2}{\sum(x-\bar{x})^2\sum(y-\bar{y})^2}$$
$$= \left\{\frac{\sum(x-\bar{x})(y-\bar{y})}{\sqrt{\sum(x-\bar{x})^2\sum(y-\bar{y})^2}}\right\}^2 = r_{xy}^2$$

So, it can be written as

$$r_{xy} = \pm\sqrt{b_{\frac{y}{x}} \times b_{\frac{x}{y}}}$$

Then, the definitive formulae can be can be applied to find the followings.

$$\frac{r_{xy}}{b_{\frac{y}{x}}} = \frac{\frac{\sigma_{xy}}{\sigma_x \sigma_y}}{\frac{\sigma_{xy}}{\sigma_x^2}} = \frac{\sigma_x}{\sigma_y}$$

$$\Rightarrow r_{xy} = b_{\frac{y}{x}} \times \frac{\sigma_x}{\sigma_y}$$

$$\Rightarrow b_{\frac{y}{x}} = r_{xy} \times \frac{\sigma_y}{\sigma_x}$$

$$\frac{r_{xy}}{b_{\frac{x}{y}}} = \frac{\frac{\sigma_{xy}}{\sigma_x \sigma_y}}{\frac{\sigma_{xy}}{\sigma_y^2}} = \frac{\sigma_y}{\sigma_x}$$

$$\Rightarrow r_{xy} = b_{\frac{x}{y}} \times \frac{\sigma_y}{\sigma_x}$$

$$\Rightarrow b_{\frac{x}{y}} = r_{xy} \times \frac{\sigma_x}{\sigma_y}$$

Rank Correlation: Sometimes there doesn't exist a marked linear relationship between two random variables but a monotonic relation (if one increases, the other also increases or instead, decreases) is clearly noticed.

If instead of measuring the correlation between two sets of continuous random variables x & y we replace their numerical values by their rankings, then we obtain the Rank Correlation coefficient. The rank of the $i-th$ element of a sample of size N is equal to the index of the order statistic.

The Rank Correlation coefficient

$$r_{rank} = 1 - \frac{6\sum d_i^2}{N(N^2-1)}\ ;\ d_i = x_i - y_i$$